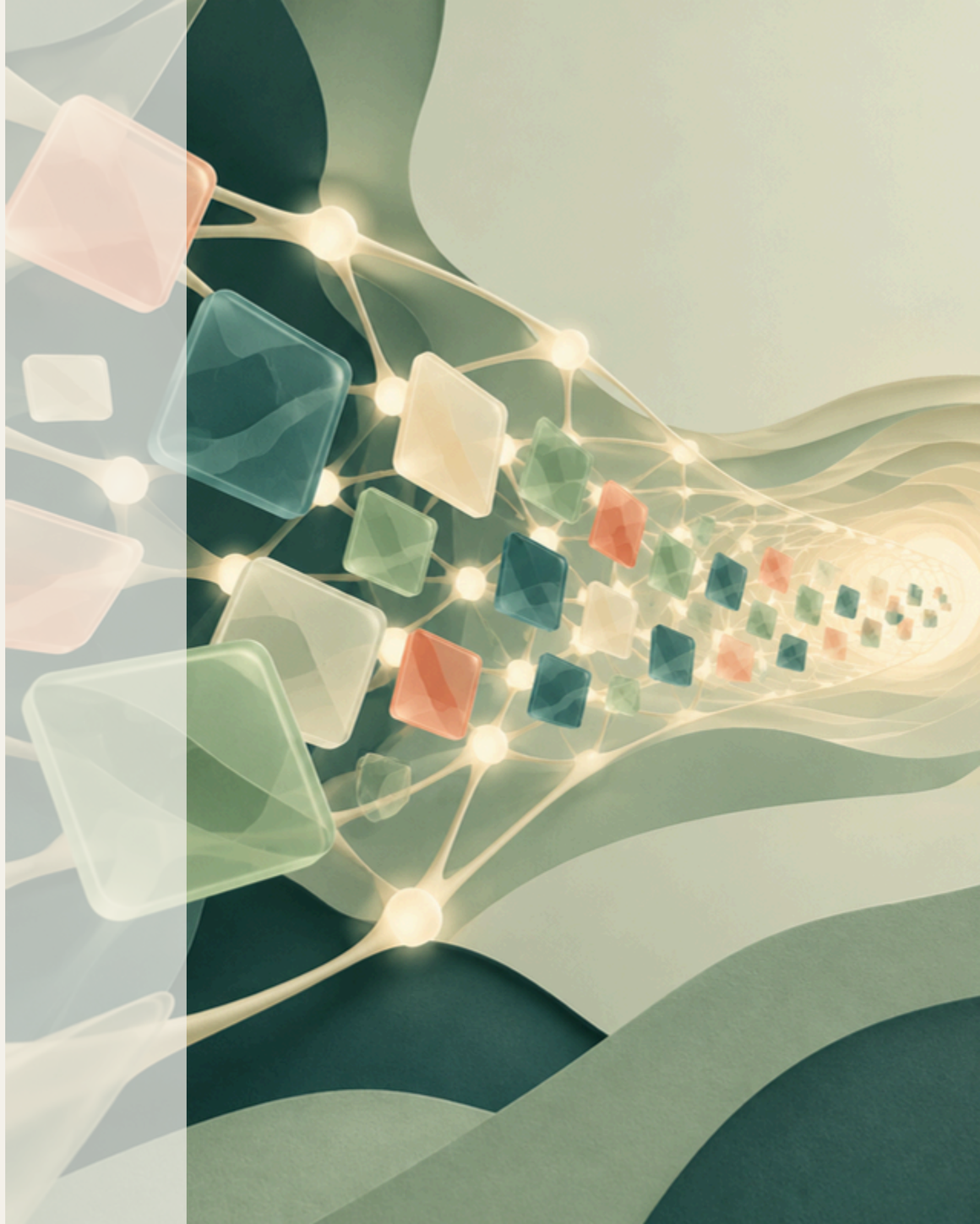
Language Models

Lecture 6 · Retake

AI for Good · Nina Borutyńska · 25044923



Poke an LLM

01

Facts about The Hague

02

Same task, 3 prompts

03

Explain it 3 ways

04

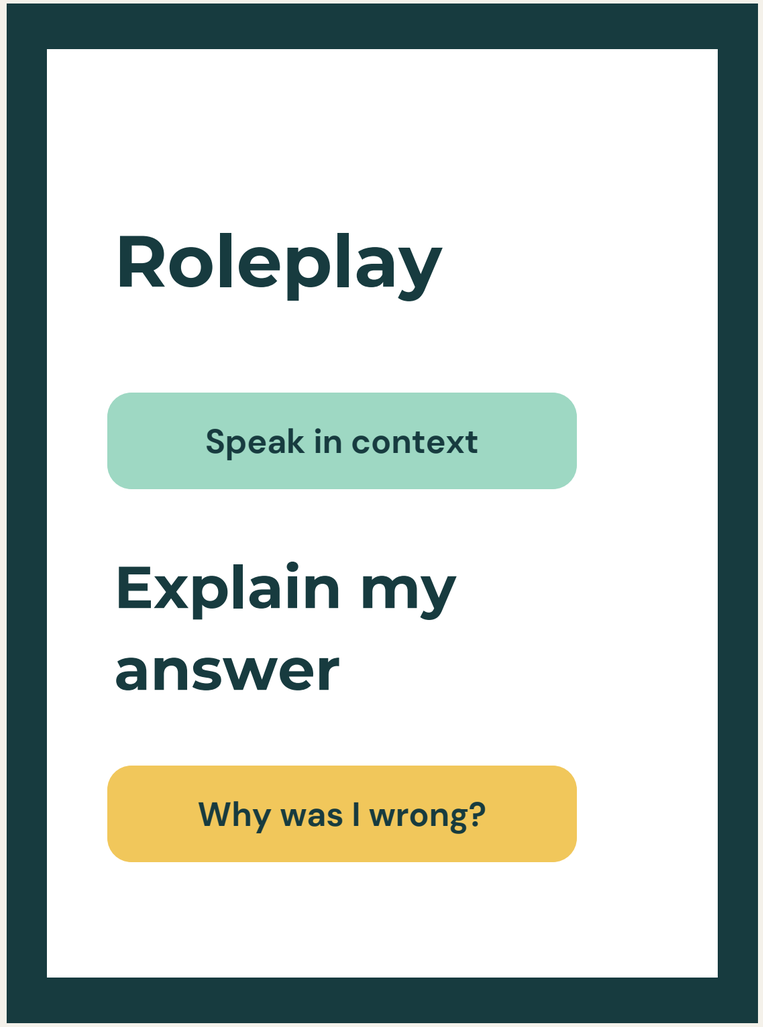
One diagnosis, please

8 districts: all correct · vague → ready to send · style changed, facts stayed · "meningitis" + call 112

An LLM-powered product

Duolingo Max

Since 2023, GPT-4
Better: explains WHY I was wrong
Concern: can be wrong too, paid plan only



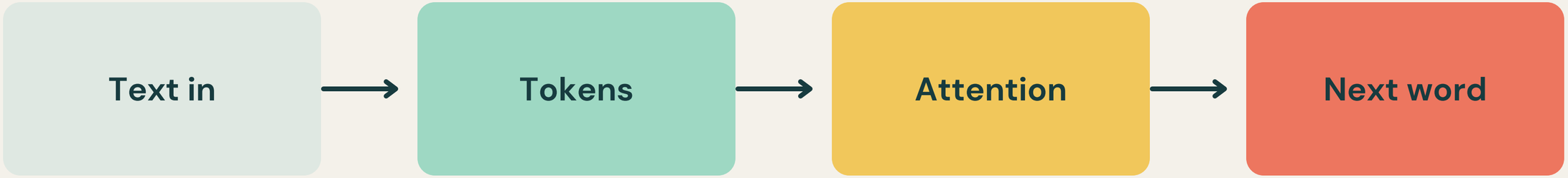
What is an LLM?

predict the next token



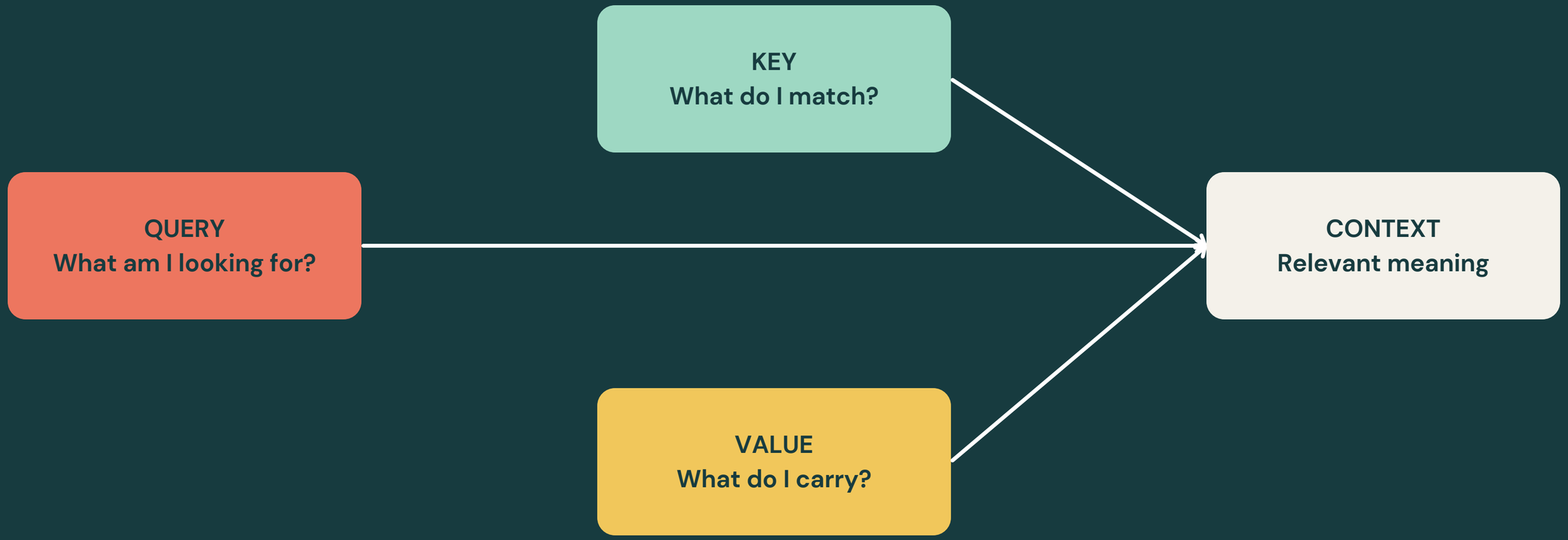
A neural network that learned one skill

The transformer in 4 steps



split → compare → weigh → predict

Attention: query, key, value



Why transformers work so well

Parallel

all words at once, fast

Long range

any word reaches any word

One goal

next token covers many tasks

Failure type: hallucination

sounds confident

is wrong

Right or wrong, it sounds the same: you can only check

Safeguards: check sources · RAG · allow "I don't know" · human review

Better prompts

BEFORE

“Write a message to my landlord about my broken heater.”



AFTER

Role · context · task format · tone

Before: generic, no name, no dates → After: real dates, contract rule, 2 time slots, signed Nina

A prompt is a specification, not a wish.

RAG, agents and multimodality

RAG
study bot that answers from course pages

AGENTS
books a doctor's appointment for you

MULTIMODAL
describes photos for blind people

answer with sources → take actions → see and hear

My verdict

Most potential: multimodality

Most concern: agents

Access for everyone vs. mistakes become real actions

Thank you

Questions?

